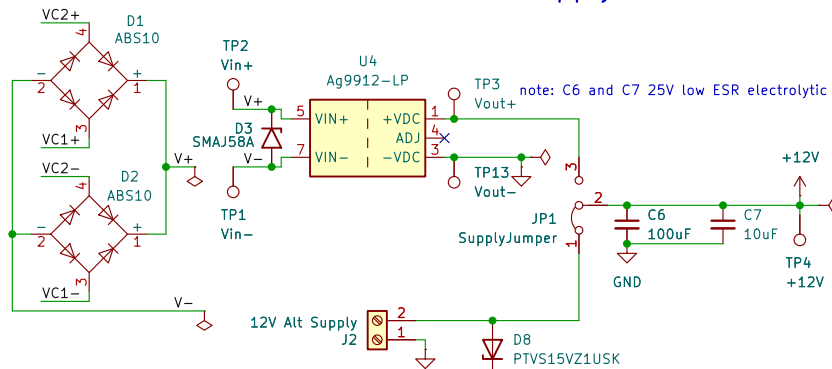


Pin connection diagram for the Raspberry Pi 4B. The diagram shows three headers: J7 (ControllerPwr), J8 (ControllerEth), and J12 (ControllerSense). The connections are as follows:

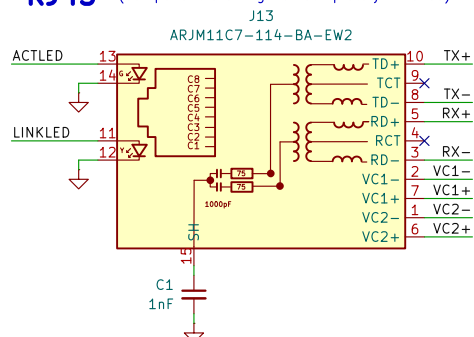
- J7 (ControllerPwr):**
 - Pin 1: +3V3D
 - Pin 2: TX+
 - Pin 3: Ground
- J8 (ControllerEth):**
 - Pin 1: TX-
 - Pin 2: TX+
 - Pin 3: RX-
 - Pin 4: RX+
 - Pin 5: LINKLED
 - Pin 6: ACTLED
- J12 (ControllerSense):**
 - Pin 7: ADC2+
 - Pin 6: ADC2-
 - Pin 5: ADC1-
 - Pin 4: ADC1+
 - Pin 3: SDI
 - Pin 2: Ground
 - Pin 1: Ground

The schematic diagram shows the signal conditioning circuit for the J10 and J11 connectors. J10 (SignalsIn) is connected to +12V and -3V3A. J11 (SignalsOut) is connected to +3V3A and ground. The circuit includes two op-amp buffers (U1, U2) and two comparators (U3, U4).

12V Supply

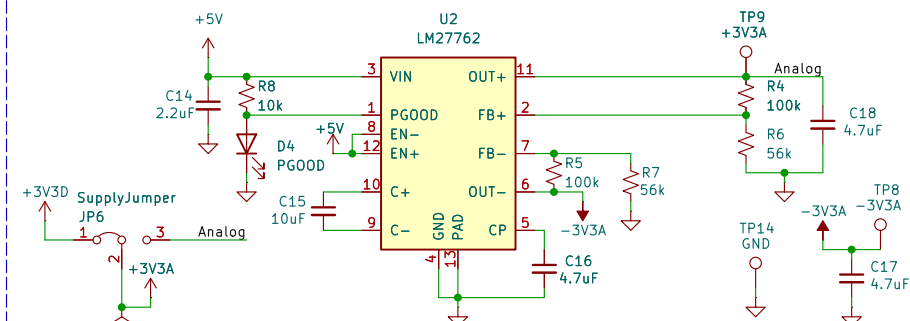


Note: Limited to 100 Base-T
(component drawing not completely finished)



Connectors and Jumpers

note: ceramic caps are recommended, X7R or X5R per datasheet



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